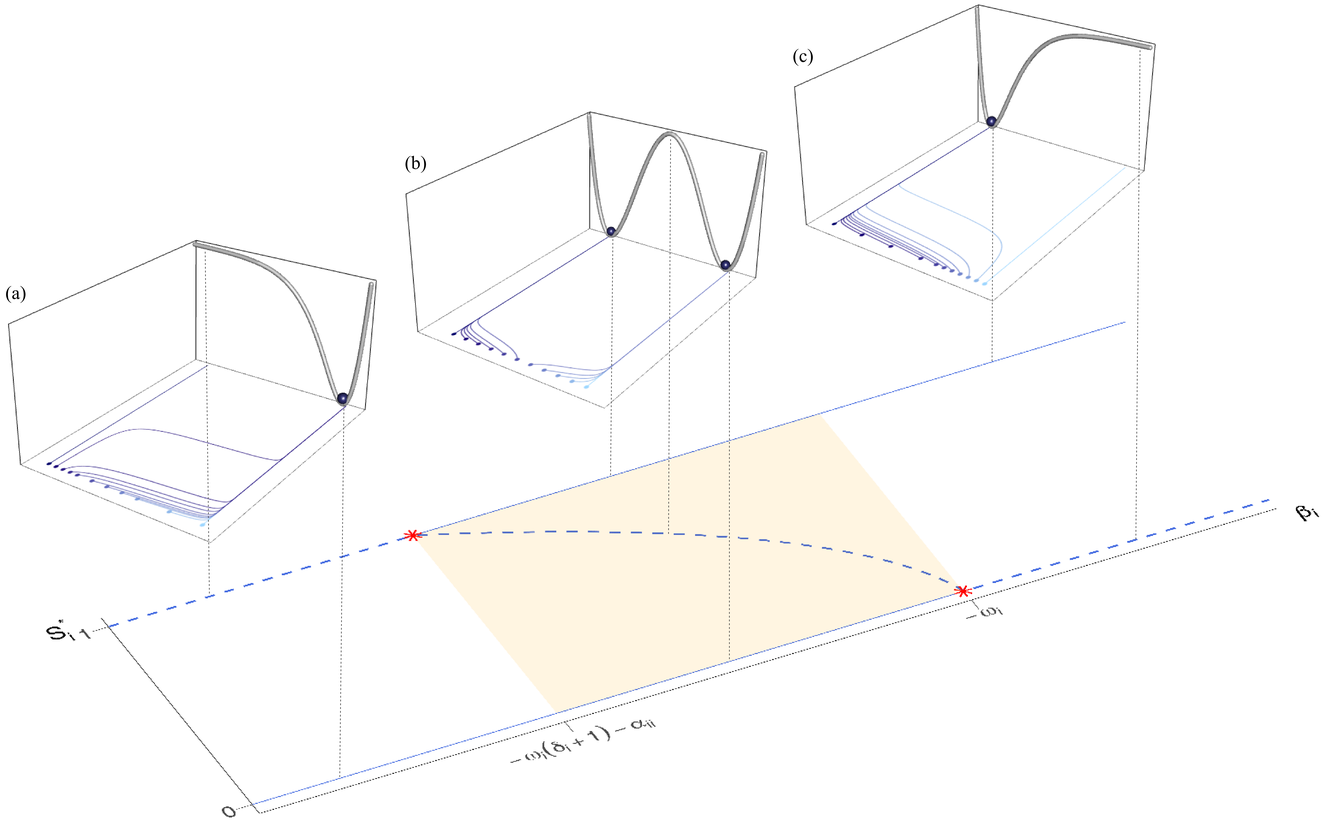
(a)
(b)
(c)
β_1
s^*_{11}
$-\omega_1$
$-\omega_1(\delta_1+1)-\sigma_{11}$
0